# Chapter 42 – Electrical Power
## Calculus Physics II

**Electric Power (P)** – is the amount of energy used/work done by the circuit per second. This energy is coming from the electric potential created by the established electric field. This energy cab be converted to thermal or mechanical energy.



$$Power\ P = \frac{dW}{dt} = \frac{dqV}{dt} = \left(\frac{dq}{dt}\right)V = IV$$

Where Electrical work = qV and $I = \frac{dq}{dt}$

The electrical power used in an electrical circuit in terms of current and voltage is given as P = IV. Using Ohm's Law the power can be derived in terms of current and resistance.

| | | |
|---|---|---|
| $P = IV$ | $P = I^2R$ | $P = V^2/R$ |

Unit's of Watts (W) 1 W = 1 J/s

- ❖ Electric power is the rate at which the electric energy is being used.

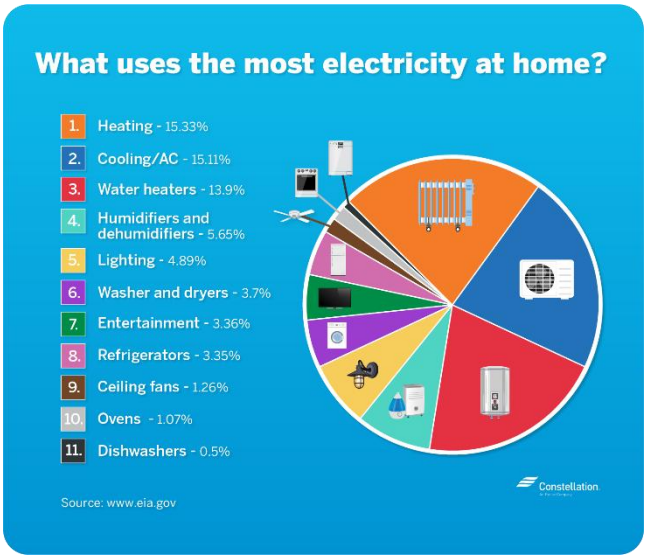
Energy Dissipation in a Resistor: The amount of energy a resistor uses or dissipates is given by

$$Energy\ used\ by\ resistor = \int Pdt$$

If the power consumed by the circuit element is constant (common for resistors) then the energy used is

$$Energy = P\Delta t$$

- A 60 W light bulb converts 60 joules of energy every second into light + heat.
- A resistor in a circuit dissipates electrical energy as thermal energy.



**Example**: Using the electrical power equations

A) If a 3 V battery can supply 0.6 A of current to run a toy car motor, how much power is needed to run the motor?

$P = IV = (0.6\ A)(3\ V) = 1.8\ W$

B) A speaker has a resistance of 8 Ω and the amplifier supplies a current through the speaker of I = 2 A, how much power is the amplifier delivering to the speaker?

$P = I^2R = (2\ A)^2(8\ \Omega) = 32\ W$

C) An electrical heater uses an element with a resistance of 18 Ω and is plug into a potential of 120 V, how much thermal energy (heat) does the heater generate if all the electrical power is converted to thermal energy?

$P = V^2/R = (120\ V)^2/(18\ \Omega) = 800\ W$

**Example**: A light has a power rating of 60 W, if the light is hooked up to a 120 V potential, what is the resistance and current flowing through the bulb?

$P = V^2/R$ therefore

$R = V^2/\ P = (120V)^2/60\ W = 240\ \Omega$

$P = IV$ therefore

$I = P/V = 60\ W/120\ V = 0.5\ A$

What would happen to the resistance and current if the 60 W bulb was replaced with a 100 W bulb?

$P = V^2/R$ therefore

$R = V^2/\ P = (120V)^2/100\ W = 144\ \Omega$

$P = IV$ therefore

$I = P/V = 60\ W/120\ V = 0.83\ A$

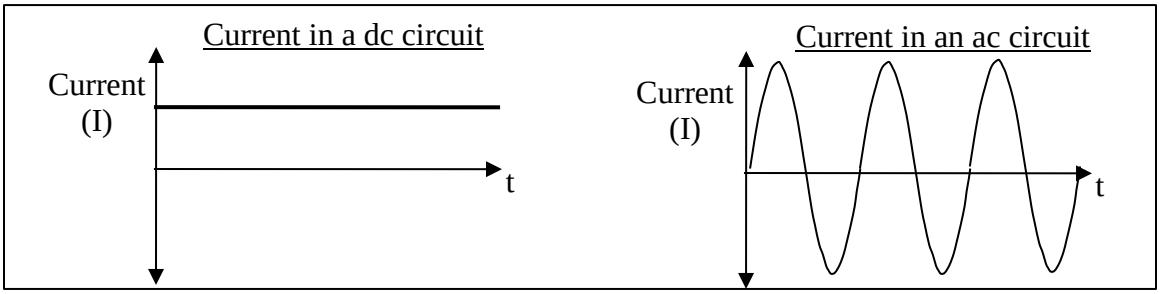
The resistance of the 100 W bulb is less than the 60 W which allows more current flow through the 100 W bulb.

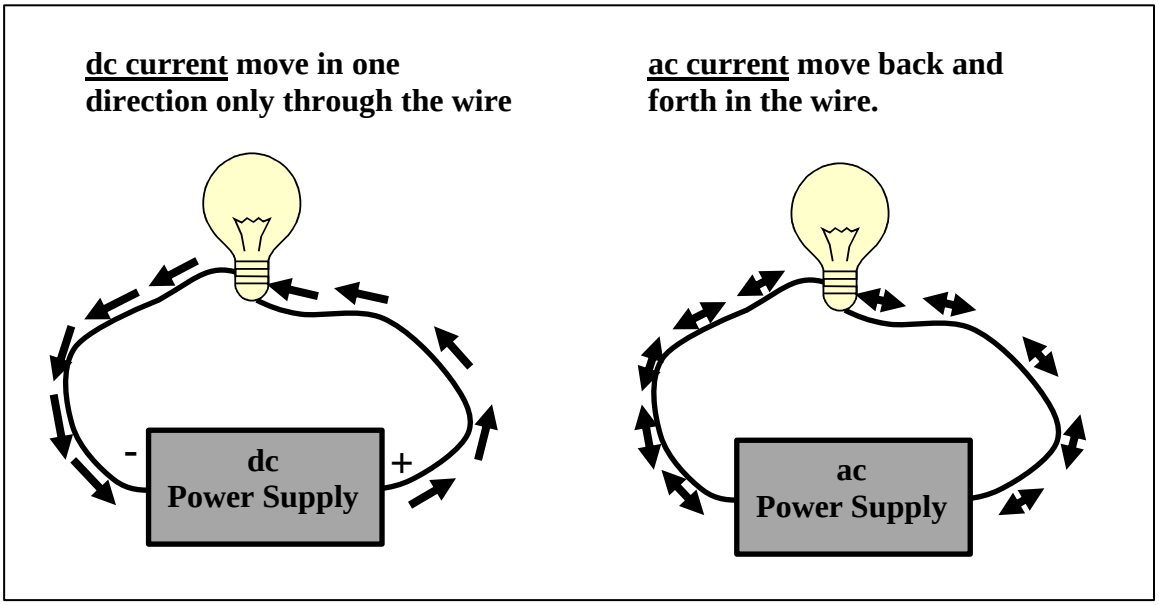
**Types of current/voltage** – There are two types of current/voltage that can be generated by a power source, direct current (dc) or alternating current (ac).

Note: We will talk about ac and dc current, but this also applies to the potential that is driving the current!

**Direct Current (dc)** – Charge moves around the circuit in the same direction at all times. The electric potential of the dc power source does not change polarity, it does not switch from positive to negative. Examples: batteries, solar cells

**Alternating Current (ac)** – The current in the circuit moves back and forth, it flows in one direction then back in the opposite direction, because the potential of the power supply changes (cycles) from positive to negative and back to positive over and over again. Examples: electric generators (power plants), function generators.

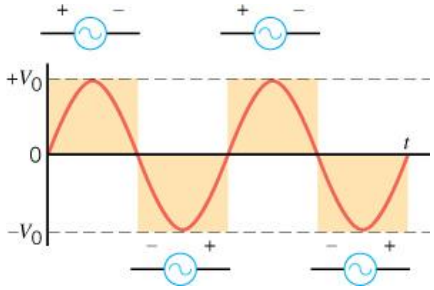



- If a circuit is Ohmic (it obeys Ohms Law) then Ohms Law can be applied the same to both dc and ac circuits.

**Describing Alternating Current** – For ac current, the circuit periodically reverses direction, it flows in one direction then back in the opposite direction. This reversing of direction of the current is due to the potential of the power supply periodically changing from positive to negative and back to positive, driving the charge back and forth in the circuit. This periodic change can be described by the "sin" function and the current and voltage are given by

$I = I_p \sin(2\pi ft)$ $V = V_p \sin(2\pi ft)$

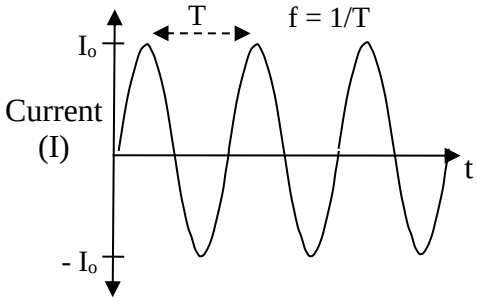



I is the current and V is the voltage at a given time (t).
$V_p$ (or $V_o$) and $I_p$ (or $I_o$) are the peak voltage and current respectively and
f is the frequency of the voltage or current in units of Hz.

The current peaks along with the voltage at the same moment in time, the current and voltage are in phase when they are generated.

- "2πft" term in the above equation tells the position along the wave. It tells where you are in the cycle in units of radians ($\Theta = \omega t = 2\pi ft$)
- For the case of most electrical devices the current flows as a sinusoidal pattern with time and can be described by the number of cycles the current goes through per second or frequency (f = 1/T)

Current in an ac circuit




**The Root Mean Square (rms) Values** – The rms value is used when describing the properties of a time varying electric signal where $I_p$ (or $I_o$) is the peak current and $V_p$ (or $V_o$)is the peak voltage. The rms value of voltage and current gives the voltage or current that would create the same power that a d.c. energy source would at that same voltage and current.

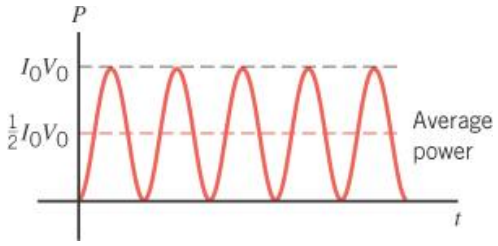
$$P_{rms} = P_{average} = \frac{1}{2} I_p V_p = \left(\frac{I_p}{\sqrt{2}}\right)\left(\frac{V_p}{\sqrt{2}}\right)$$

$P_{rms} = P_{d.c.}$

$$I_{rms} = \frac{I_p}{\sqrt{2}} \qquad V_{rms} = \frac{V_p}{\sqrt{2}}$$

$P_{rms} = I_{rms} V_{rms} = I^2_{rms} R = V^2_{rms} / R$

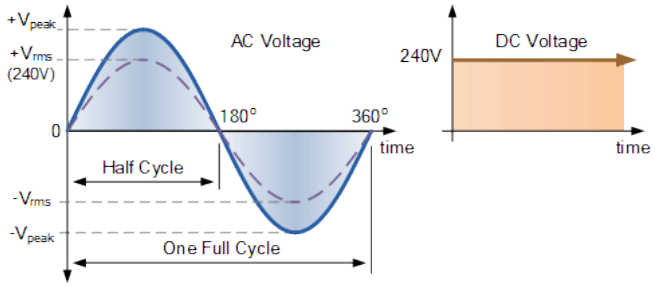
$V_{rms} = I_{rms} R$ $P_{avg} = ½ I_p V_p$




- The rms values are used to specify most ac circuits.
- Home voltage is specified as 120 $V_{rms}$ but the voltage has a peak value of 170 V.

**Important:** The rms value of current results in the same electric power as does the d.c. value of current. A resistor or light bulb would have the exact same power delivered to it by 10 A of rms a.c. current as that of 10 A of d.c. current. $P_{rms} = P_{d.c.}$

**Example:** The graph shows an a.c. voltage with a peak voltage ($V_p$) of about 340 $V_{peak}$ and an rms voltage of 240 $V_{rms}$. This same rms voltage (240 $V_{rms}$) will deliver the same amount of power that the d.c. voltage of 240 $V_{d.c.}$ shown in the second graph.

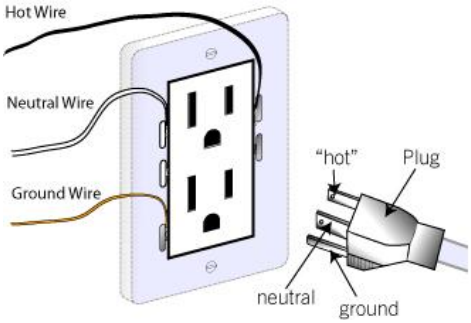



**Example:** If your home ac voltage is 120 $V_{rms}$, what is the peak voltage?

$$V_{rms} = \frac{V_p}{\sqrt{2}} \text{ therefore } V_p = \sqrt{2}[V_{rms}] = 170 \text{ V peak}$$

## Household Electricity and Safety

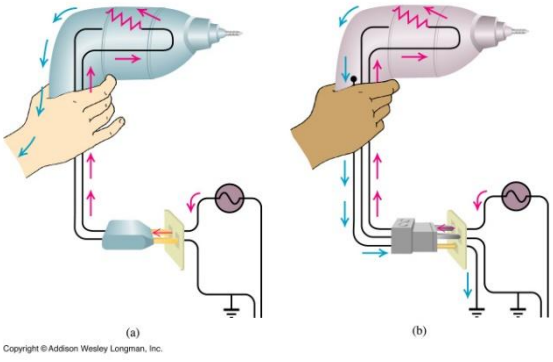
**The Electrical outlet** - The electrical power supplied to a home is in the form of an a.c. voltage supplied by the electric grid. The electrical plugs that are used in a house use three wires, Hot (black wire), the Neutral (white wire) and the ground (green wire). When an appliance is plugged into the wall outlet the current flows between the Hot wire and the Neutral wire. The ground wire is for safety.




**Electrical Grounding** - There are two main reasons to have an electrical ground

1) **Same Potential Reference** – The ground is connected to the "Earth" therefore the ground is the same for everybody and the ground is at zero potential. The potential different is measured from the same point for everybody therefore every device "sees" the same voltage.

2) **Safety** – The ground removes any excess charge on conductors so that the charge does not go through a person to ground but goes through the grounding wire.



Example of a drill – The drill in the first image (a) does not have a grounding wire, only hot and neutral so when the wire shorts out on the drill case there is a conduction path through the arm down the body to ground (NOT a good thing!).

But for the second image (b) the case is attached to a grounding wire therefore when the wire shorts to the case the charge (current) goes to ground through the grounding wire not the person holding the drill. (Good thing!)

**The Breaker or Fuse** – Breakers or fuses are used so the flow of current does not exceed the safety limits of the wire, if more current flows through the wire than should the breaker trips and creates an open circuit so the current stops flowing. If too much current is allowed to flow in a wire the wire could heat up and this may cause a fire or create a dangerous situation for a person that could come in contact with the wire. Also, if too much current is flowing then there could be a "short circuit" and breaker would stop the flow of current to the "short".

**Electricity and the human body** – when the human body comes in contact with a potential difference (voltage) current can flow through the body. The amount of current depends on where on the body the contact is made and the condition of the skin (wet, dry, etc. ,.,).

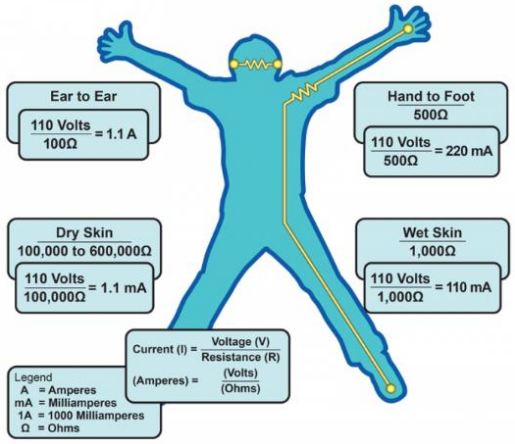
- It is the voltage that drives the current through the body, but it is the current that causes the problems. A few milliamps through the body and you feel a slight tingling sensation but over 100 mA and it could be fatal!

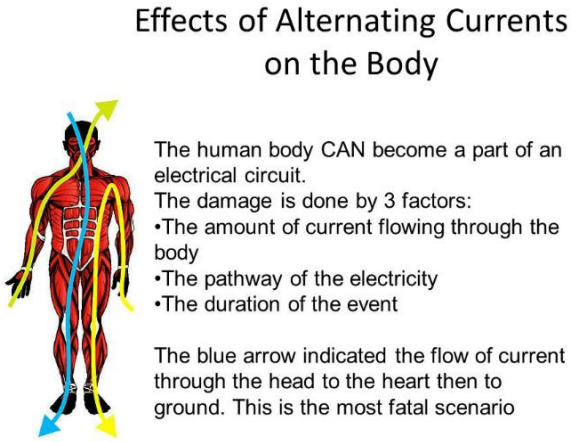
| CONDITION | EFFECTS |
|---|---|
| 1-3mA of current | Mild sensation |
| 10mA of current | Muscles contract, releasing grip may be difficult |
| 30mA of current | Breathing difficult, possible loss of consciousness |
| 30-75mA of current | Respiratory paralysis |
| 100-200mA of current | Ventricular fibrillation |
| 50-300mA of current | Shock (potentially fatal) |
| Over 1500mA of current | Tissue and organ burn |
| 150° F | Cell destruction |
| 200° F | Skin experiences "third degree" burns |

**Electrical Shock** – Current passing through the body by way of a closed circuit. The current can flow from conductor to conductor or from conductor to ground. The current (electric charge) interferes with the normal functioning of the body's nervous systems (sensations and muscle contraction) or burns the tissue.

- The amount of current that will flow though the body depends on many factors: Voltage of the source, path that the current takes through the body, moisture content of skin, etc.

- The diagram shows approximate values for the resistance of certain body parts for current flow along with the amount of current that could flow




**Tips for Electrical safety basics** Basic Electrical Safety | EHS (washington.edu)

- Don't work with exposed conductors carrying 50 volts or more.
- Make sure electrical equipment is properly connected, grounded and in good working order.
- Extension cords may not be used as permanent wiring and should be removed after temporary use for an activity or event.
- High amperage equipment such as space heaters, portable air conditioners and other equipment must be plugged directly into permanent wall receptacles.
- Do not access, use or alter any building's electrical service, including circuit breaker panels, unless you are specifically qualified and authorized to do so.
- Wet environments can increase the risk of an electrical shock.
- Inspect cords and plugs regularly for signs of wear or exposed wires.
- Look for fraying, exposed wires, or loose connections. Replace damaged cords immediately.
- Avoid overloading outlets - Plugging too many high-watt devices into a single outlet or power strip can cause overheating and fires.
- Keep electrical devices away from water.
- Use the right gage extension cord for the device being connected.
- Unplug appliances by the plug, not the cord. Pulling the cord can damage internal wiring and create a shock hazard.
- Install ground-fault circuit interrupters (GFCIs). Use GFCI outlets in bathrooms, kitchens, garages, and outdoor spaces to prevent shock.
- Turn off power before repairs. Always shut off the circuit at the breaker box before performing electrical work.
- Hire a licensed electrician for major work. DIY electrical modifications can be dangerous; a professional ensures safety and code compliance.